



अश्वाMOBILITY

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SIMULATION
TO SPEED.

ASHWA RACING

**JULY
2025
EDITION**

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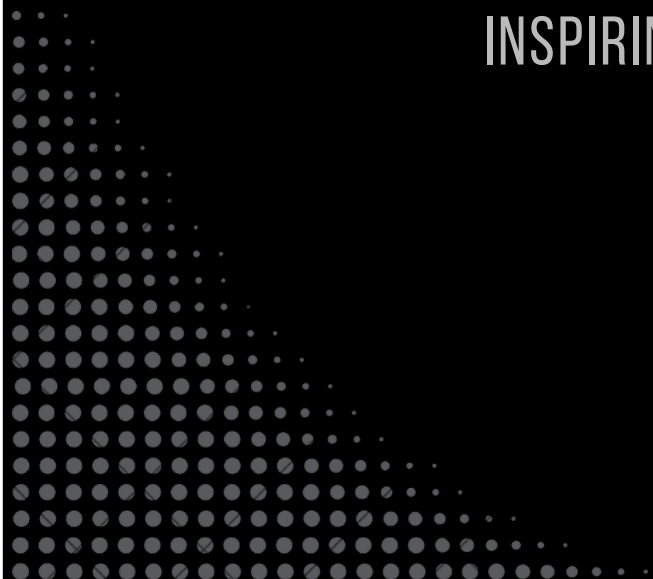


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**VISION
AND
MISSION**

GUIDING PURPOSE,
INSPIRING FUTURE.



> MISSION AND VISION <

VISION

- To be the national leader in all categories of Formula Student, to push the limits of engineering and compete in events at the highest level.
- To foster student mobility solutions in the future by developing sustainable prototypes across various domains.

MISSION

- Prepare students for the industry by designing and building world-class vehicles while facilitating industry interaction.
- Showcase India's engineering expertise in mobility on a global stage and provide talent opportunities for the industry.
- Develop cost-effective, sustainable transportation solutions for the masses.

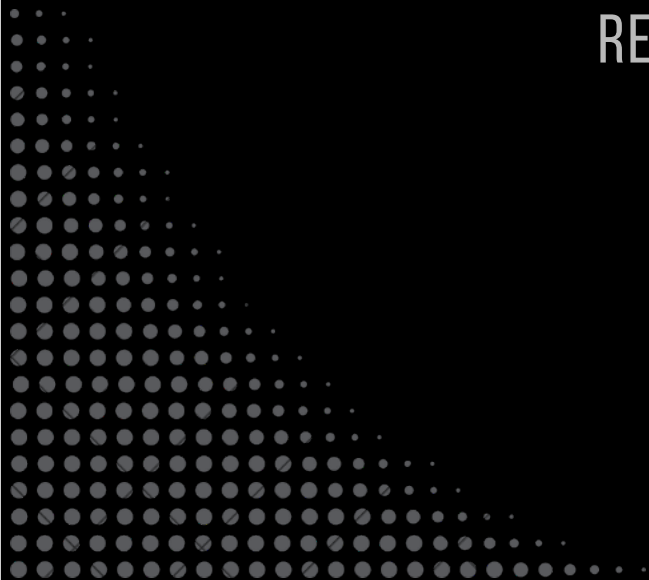


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**FROM
CAD TO
TRACK**

THE DESIGN
REVIEW.





FROM CAD TO TRACK: THE DESIGN REVIEW.

The first design review of the season was successfully conducted, marking a major milestone in our project timeline. Each subsystem—Drivetrain, Suspension, Chassis, Engine, Brakes, and Electrical—presented their design progress, covering key areas such as material selection, calculations and simulations, CAD models, sub-assembly development, rule compliance, and considerations for manufacturability and serviceability.

The review followed a detailed evaluation rubric, with each criterion assessed on a scale from Unacceptable to Excellent. This structured format ensured consistency and helped highlight both strengths and areas needing improvement.

All team leads and subsystem members participated actively, and valuable feedback was collected throughout the day. Within 24 hours, the Chief Engineer categorized the feedback into three actionable levels—Must Fix, Need to Fix, and Nice to Have—setting the stage for the next phase of refinement. The review not only helped us assess technical readiness but also brought the entire team one step closer to a cohesive and production-ready design.



Fig. Subsystems presenting design models and progress during the design review.

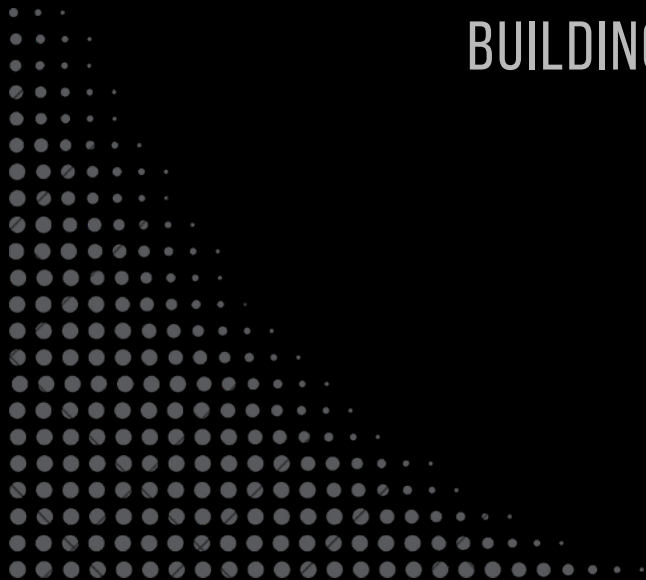


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ALUMNI CORNER

CONNECTING MEMORIES,
BUILDING FUTURES!



ALUMNI CORNER



Manasi Kadambi, Partnerships, Marketing and PR, 2020

My journey with Ashwa Racing:

My journey with Ashwa Racing began in 2017 when I joined the Sponsorship, Marketing & PR team. As a Biotechnology major, it was a bold step into unfamiliar territory—but one that quickly became a defining chapter in my undergrad life. Over the next three years, I worked on everything from reaching out to potential sponsors and closing deals, to showcasing at high-visibility events like IMTEX and managing our brand presence. It wasn't just about raising funds—it was about telling our story, creating value, and ensuring the team had what it needed to build the car we all believed in. Ashwa pushed me beyond my comfort zone and sparked my love for tech, strategy, and the power of great execution.

My competition experience:

As part of the 2020 team, I had the chance to compete at two incredible events—Formula Hybrid+Electric 2019 in the U.S. and Formula Bharat 2020. For FH+E, I led sponsorship efforts and experienced my first international engineering competition. It gave me perspective—not just on tech, but on how teams across the world think, innovate, and collaborate. At Formula Bharat, I delivered our Business Plan Presentation and we placed 3rd. That experience, both in preparation and in the pitch room, shaped how I think about product, value propositions, and storytelling even today.

How Ashwa influenced my growth as an engineer:

Ashwa taught me that great engineering and innovation don't exist in silos—they need structure, planning, and strong cross-functional communication to make an actual impact. I learned early on that solving real-world problems is not just about building the right solution, it's about identifying the right problem, aligning with stakeholders, and making sure your solution makes business sense. That foundational understanding is what led me to my first role in product marketing at a fintech firm, and eventually shaped my path toward product management.

Standout projects or learnings:

Working on the Business Plan Presentation for Formula Bharat was a game-changer. It brought together everything I loved: strategy, creativity, business thinking, and clear communication. That single project played a major role in landing my first job and set the tone for everything that followed. I went on to pursue my Master's in Engineering Management at Northeastern University with a specialization in product management and tech innovation. I did my co-op with Honeywell Intelligrated in 2023 and have now joined their Offering Management team full-time. From biotech to fintech to industrial automation—it's been a winding road, but what's helped me stay grounded is the ability to learn fast, solve problems deeply, and adapt my skills across industries.

How the team shaped my personal and professional development:

Ashwa was a crash course in teamwork, resilience, and growth. It was the first time I worked so closely with people from different disciplines, all united by the shared mission of building the car. We had high highs and tough lows (FB 2020 didn't go as planned), but I learned that failure isn't the end of the story. What matters more is how you reflect, realign, and keep moving forward. Being one of the very few women on the team, I also understood early on that showing up mattered—being confident, competent, and consistent was key. It wasn't about fighting the status quo, it was about showing what's possible when you rise above it.

Closing note:

The Formula Student community has given me more than I could've imagined, and it's something I've been proud to give back to. I've served as a judge for the Business Plan Presentation at Formula Bharat since 2021 (now leading the event), and at Formula Hybrid+Electric from 2022 to 2024. Every time I interact with young, driven teams, I'm reminded of what Ashwa gave me—clarity, direction, and a tribe of people who want to build, challenge, and grow. For that, I'll always be grateful.



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THROTTLE TALK

DECODING THE ELECTRONIC
THROTTLE SYSTEM





THROTTLE TALK: DECODING THE ELECTRONIC THROTTLE SYSTEM

Ever wondered how your bike's quickshifter works? Or how switching between rain, sport, and eco modes completely transforms the feel of your ride? Or how traction control steps in to save countless riders from a crash? All of this is made possible thanks to advancements in the Electronic Throttle Body (ETB) system.

At Ashwa Racing, we have been using the Electronic Throttle Body (ETB) system since 2024, enabling the implementation of advanced features such as quickshifter and auto-blipper.



Fig. The accelerator pedal with a 6 pin connector for the apps housing two rotary position sensors.

The primary components of the Electronic Throttle System include the Accelerator Pedal Position Sensor (APPS) and the Electronic Throttle Body itself, which contains the butterfly valve responsible for regulating the airflow into the engine.

We use an APPS from Williams Controls, which is attached to the accelerator pedal housing. It consists of two inductive rotary position sensors along with a signal conditioning circuit that outputs the positions of both independent sensors. These signals are sent to the MoTeC M800 ECU, which then calculates the target position of the butterfly valve in the ETB.

The MoTeC ECU sends a PWM signal to the ETB's stepper motor, which actuates the butterfly valve accordingly. The ETB also includes two Throttle Position Sensors (TPS) that continuously measure the actual position of the valve and feed this information back to the ECU for closed-loop control.

Now that the basics of the Electronic Throttle System are out of the way, here comes the fun part! Using the MoTeC M800 ECU, we can detect when the upshift button is pressed and momentarily cut power by interrupting ignition or fuel for a short duration. This allows seamless clutchless upshifting on our sequential gearbox.

When the downshift button is pressed, the ECU uses the ETB to automatically blip the throttle, momentarily increasing engine RPM to match the gearbox input speed. This rev-matching technique allows smooth, clutchless downshifting.

The Electronic Throttle Body (ETB) also enables precise and stable idle control through a closed-loop PID (Proportional-Integral-Derivative) algorithm. By allowing fine-tuning of each term—proportional, integral, and derivative—within the MoTeC M800 ECU, we can accurately manage the throttle opening to maintain a consistent idle speed under varying engine loads and conditions. The ETB's ability to make real-time, ECU-controlled adjustments eliminates the need for mechanical idle valves, simplifying the system while improving its responsiveness and reliability.

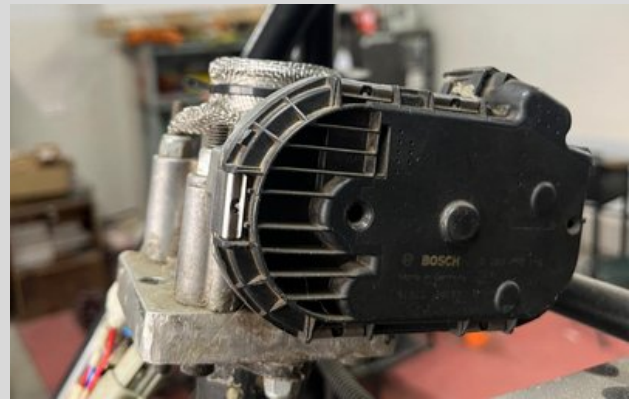


Fig. The Bosch ETB housing the butterfly valve and the stepper motor.

Safety, Served by Wire

To ensure the safety of the ETB system, multiple layers of hardware and software checks are implemented, starting with the use of redundant sensors. Both the Accelerator Pedal Position Sensor (APPS) and the Throttle Position Sensor (TPS) are equipped with dual independent channels. This redundancy allows the ECU to constantly verify the plausibility of the signals by comparing the two channels for consistency.

In the case of the APPS, we use a dedicated APPS board that monitors both channels in real time. If the two signals differ by more than 10%, the board flags this as a fault and triggers a safety mechanism—cutting power to the ETB to prevent any unintended throttle movement. A similar check is done for the TPS, comparing the feedback signals from the throttle body to ensure that the actual throttle position matches the commanded position within the same 10% tolerance.

Additionally, since the APPS and TPS signals are Single Channel Sensor (SCS) types, the system includes short-circuit checks. On the XX5C which was the 2025 prototype, an arduino nano is used to monitor the signal lines for unexpected voltage levels that could indicate a short to ground or supply. If a fault is detected, these circuits immediately trigger the Shutdown Circuit (SDC) which cuts power to the fuel injector, fuel pump and ignition.

Together, these checks form a robust fail-safe architecture that allows the ETB system to operate safely and reliably even in the demanding conditions of a race environment.

These integrated checks create a strong, fail-safe architecture, enabling the ETB system to function safely and dependably, even under the rigorous demands of a race environment.

*Akshay Urs,
Electrical Member,
'27 Batch Ashwa Racing.*



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(2025-2026)**

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Ansys



AKSHAYA MOTORS





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**TEAM
DIRECTORY
2026**

DRIVEN BY PASSION, UNITED BY ENGINEERING—
THIS IS ASHWA RACING!

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